

Introduction 修改建议：机器人损坏与在线重规划

针对当前 LTL-MRTA 论文 Introduction 中“实际实验中机器人损坏，因此进行在线重规划”的写法建议

一、总判断

建议把“机器人损坏”写成实际执行中的 robot unavailability / available robot set changes，并把“在线重规划”写成 online repair / replanning of the remaining mission from the current execution state。它应该作为工程动机和实验补充出现，不建议单独升级为新的理论创新点。

原因是：如果把它写成核心创新点，审稿人会自然追问故障检测模型、故障发生时刻、多个机器人同时失效、通信中断与损坏的区别、重规划完备性/最优性证明以及系统化故障实验。如果目前论文的重点仍然是 automaton generation 中嵌入 planning、GBA/NBA 阶段 pruning、plan graph 与 warm-start BnB，那么机器人损坏应当只作为执行阶段的在线修复能力来呈现。

二、当前 Introduction 中最适合插入的位置

位置	当前上下文	修改目的
Introduction 第二段中“decisions may need to be updated online”之后	当前已经在讲 large-scale LTL-MRTA 中不仅要快速得到可执行初始解，还可能需要在在线更新决策。	补充真实实验中机器人损坏、定位失败、电量耗尽等导致可用机器人集合变化，从而需要从当前状态修复剩余任务。
Related Work 末尾总体定位段	当前已经在总结你的方法：GBA 剪枝、NBA 与 plan graph 同步生成、warm-start BnB，并已有一句 “When robot failures occur ...”。	把这句改得更克制、更自然，避免显得突然或过度声称“解决机器人故障问题”。
Contributions 段最后一句或实验说明处	当前贡献主要是框架、剪枝/plan graph、warm-start BnB。	只轻描淡写地说明 real-robot experiment 包含 robot-unavailability case，不新增 contribution 条目。

三、插入位置 1：Introduction 第二段中补充工程动机

建议放在下面这句话之后：

当前原文定位句

In practical robotic applications, however, it is often important to obtain an executable solution early, especially in large-scale settings where task execution needs to start promptly and decisions may need to be updated online.

建议插入以下英文段落：

推荐插入文本

In real-robot execution, the available team may also change unexpectedly due to hardware faults, robot damage, localization failures, or battery depletion. Such events do not necessarily change the temporal-logic mission itself, but they invalidate assignments involving the unavailable robots and require the remaining mission to be repaired from the current execution state. Therefore, an LTL-MRTA planner should not only generate an executable plan before execution, but also support fast online replanning when the available robot set changes.

这段的作用是把机器人损坏与在线重规划放进“实际执行约束”的语境中，而不是把它写成新的方法模块。这里用 `available team changes` 和 `unavailable robots` 比反复使用 `robot damage` 更稳妥，因为它覆盖硬件损坏、定位失败、电池耗尽、通信或执行异常等常见真实实验情况。

四、插入位置 2: Related Work 末尾定位段的替换写法

你当前末尾已有 “When robot failures occur during execution, the proposed method can rapidly repair the plan for the remaining mission.” 这句话，但它前面铺垫不足，容易让审稿人误解为新的 robustness contribution。建议把它并入方法定位段，写成“执行阶段可用机器人集合变化时，从当前执行状态重规划剩余任务”。

默认推荐使用下面的保守版；只有当代码确实复用旧的 `plan graph / warm-start structure` 时，才使用后面的加强版。

保守版：如果在线重规划是从当前状态重新调用规划流程

During GBA construction, infeasible and decomposition-inducing transitions are pruned before they are propagated to the NBA stage. During NBA construction, the NBA and a shared planning graph are generated synchronously, so that an accepting path of the NBA can yield an executable initial plan as soon as it becomes available. The resulting plan graph then provides a warm start for a branch-and-bound assignment optimizer, which continues to improve the makespan. Moreover, the same planning workflow can be invoked during execution: when a robot becomes unavailable in real experiments, the executed prefix is fixed, the failed robot is removed from the available set, and the remaining mission is replanned from the current team state. The proposed framework therefore supports fast initial planning, online repair under robot unavailability, and continued improvement toward higher-quality solutions.

加强版：只有在确实复用 `planning structure` 时使用

During GBA construction, infeasible and decomposition-inducing transitions are pruned before they are propagated to the NBA stage. During NBA construction, the NBA and a shared planning graph are generated synchronously, so that an accepting path of the NBA can yield an executable initial plan as soon as it becomes available. The resulting plan graph then provides a warm start for a branch-and-bound assignment optimizer, which continues to improve the makespan. Moreover, the generated planning structure can be reused during execution: when a robot becomes unavailable in real experiments, the executed prefix is fixed, the failed robot is removed from the available set, and the remaining mission is replanned from the current team state. The proposed framework therefore supports fast initial planning, online repair under robot unavailability, and continued improvement toward higher-quality solutions.

五、Contributions 段的最小补充

不建议新增一条 contribution。可以在 contributions 后的实验总结句中加入一句，作为实验验证内容：

贡献段末尾推荐改写

Experimental results demonstrate the effectiveness of the proposed method, especially in complex tasks and large-scale problem instances. The real-robot experiments further include a robot-unavailability case, showing that the proposed planning workflow can repair the remaining mission online during execution.

如果你想稍微强调 `plan graph` 的作用，可以改成：

贡献段末尾加强版

Experimental results demonstrate the effectiveness of the proposed method, especially in complex tasks and large-scale problem instances. The real-robot experiments further include a robot-unavailability case, showing that the generated planning structure can support online repair of the remaining mission during execution.

六、写作边界：哪些说法不建议使用

不建议写法	问题	推荐替换
To address robot damage, we propose a resilient online replanning algorithm.	这会把机器人损坏提升为新的核心创新点，容易引出故障模型、鲁棒性证明和系统故障实验要求。	The planner supports online repair when the available robot set changes during execution.
The proposed method is robust to robot failures.	“robust to failures” 声称太强，除非有严格 failure model 和大量故障实验。	The proposed planning workflow can repair the remaining mission after robot unavailability is detected.
Robot damage is solved by our framework.	过于工程化且不够学术，也暗示问题被完整解决。	Robot unavailability is handled by fixing the executed prefix, removing the unavailable robot, and replanning the remaining mission.
resilient online replanning as a contribution	如果不是核心理论模块，不应作为单独 contribution。	execution-level online repair / robot-unavailability case in experiments

七、建议使用的术语

- robot unavailability: 最推荐。比 robot damage 更宽泛、更学术，也不会过度绑定具体损坏原因。
- available robot set changes: 适合放在 problem motivation 中，强调在线重规划的触发条件。
- online repair: 比 online replanning 更偏“修复已有计划”，更贴合执行中机器人失效后的剩余任务修复。
- remaining mission: 强调任务公式本身不一定改变，只是当前已执行前缀固定，剩余任务需要重新分配。
- current execution state / current team state: 强调不是从初始状态重新规划，而是基于当前执行进度和机器人状态。

八、建议在正文中隐含的在线重规划逻辑

Introduction 中不需要展开算法细节，但语义上应与正文保持一致。建议你在方法或实验部分用如下逻辑描述在线重规划：

- 检测到机器人失效或不可用后，更新可用机器人集合：删除 failed/unavailable robot。
- 固定已经执行的任务前缀，即 executed prefix 不再改变。
- 根据当前机器人位置、完成时间和已完成任务状态，构造新的 replanning initial state。
- 对 remaining mission 重新生成或修复计划，并用新的 plan 继续执行。
- 如果实际实现支持复用已有 plan graph 或 warm-start BnB，可以在方法部分明确说明；如果没有复用，不要在 Introduction 中写 reuse。

九、最终最小改动方案

如果只希望对 Introduction 做最小修改，建议只改三处：

- 在 Introduction 第二段 “decisions may need to be updated online” 后加入真实执行中机器人损坏/不可用导致在线修复的动机段。
- 将 Related Work 末尾原有 “When robot failures occur ...” 改为更克制的在线修复表述，并与 GBA/NBA/plan graph/warm-start BnB 的方法定位连在一起。
- 在 Contributions 段末尾实验总结中补一句 real-robot robot-unavailability case, 不新增 contribution 条目。

十、直接可粘贴的最终组合版本

下面给出可以直接用于论文的最小修改文本。

插入段落

```
% Insert after the sentence ending with ``decisions may need to be updated online.``
In real-robot execution, the available team may also change unexpectedly due to hardware
faults, robot damage, localization failures, or battery depletion. Such events do not
necessarily change the temporal-logic mission itself, but they invalidate assignments
involving the unavailable robots and require the remaining mission to be repaired from
the current execution state. Therefore, an LTL-MRTA planner should not only generate an
executable plan before execution, but also support fast online replanning when the
available robot set changes.
```

Related Work 末尾替换段落

```
% Replace the last positioning sentences in Related Work.
During GBA construction, infeasible and decomposition-inducing transitions are pruned before
they are propagated to the NBA stage. During NBA construction, the NBA and a shared
planning graph are generated synchronously, so that an accepting path of the NBA can
yield an executable initial plan as soon as it becomes available. The resulting plan
graph then provides a warm start for a branch-and-bound assignment optimizer, which
continues to improve the makespan. Moreover, the same planning workflow can be invoked
during execution: when a robot becomes unavailable in real experiments, the executed
prefix is fixed, the failed robot is removed from the available set, and the remaining
mission is replanned from the current team state. The proposed framework therefore
supports fast initial planning, online repair under robot unavailability, and continued
improvement toward higher-quality solutions.
```

Contributions 末尾补充

```
% Add to the final sentence of the Contributions paragraph.
Experimental results demonstrate the effectiveness of the proposed method, especially in
complex tasks and large-scale problem instances. The real-robot experiments further
include a robot-unavailability case, showing that the proposed planning workflow can
repair the remaining mission online during execution.
```

最终效果：机器人损坏被自然解释为真实实验中的可用机器人集合变化；在线重规划被解释为从当前执行状态修复剩余任务；论文主创新仍然集中在 automaton-generation-aware planning、GBA/NBA pruning、plan graph 和 warm-start BnB 上。

依据当前稿件的对应位置

当前稿件 Introduction 第 1 页已经提出 “task execution needs to start promptly and decisions may need to be updated online” 的语境；Related Work 末尾已有机器人故障修复一句，但需要改得更自然；Plan graph 章节已经说明 plan node 记录任务进度、机器人分配、位置和完成时间，且 plan graph 可在线提取 feasible plan 并作为 BnB warm start。这些内容共同支撑把机器人损坏写成执行阶段 online repair，而不是新的理论创新点。